

Trainings ▾

General Information

Centinel™ Oils, Filters, and Blend Rates

The Centinel™ systems that have a control module separate from the engine ECM are designed to be calibrated for two types of oil:

- Advanced calibration settings reduce total oil consumption when set for using Valvoline® Premium Blue® 2000 or equivalent synthetic or synthetic-blended API CH-4 engine oil. The advanced setting is **not** approved for heavy-duty engines with 1999 and later calibrations..
 - Advanced calibration is also referred to as 0.25-percent blend rate. This rate uses 1 gallon of oil for each 400 gallons of fuel consumed ($1/400 = 0.25$ -percent).
- Standard calibration is for premium mineral oils like Valvoline® Premium Blue® or equivalent API CG-4 engine oil.
 - Standard calibration is also referred to as 0.33-percent blend rate. This rate uses 1 gallon of oil for each 300 gallons of fuel consumed ($1/300 = 0.33$ -percent).

For CELECT™ and CELECT™ Plus engines, the Centinel™ control module uses calibration plugs which are provided with the system to determine the oil blend rate. For Centinel™ systems that use a CM400 control module, the control module is programmed and does **not** use calibration plugs.

For Centinel™ systems used on ISM, ISX, QSX15, QST30, QSK45, and QSK60 engines, the oil blend rates are fixed in the engine software and are **not** adjustable. These engines do **not** use calibration plugs.

Refer to Table 1 for engine-specific blend rates.

Refer to the appropriate engine Operation and Maintenance manual for the oil drain and filter change intervals for the various engine families. Table 1 is categorized by engine family, oil grade, and Centinel™ blend ratio.

Table 1, Oil Blend Rates with Centinel™ System

Engine	Centinel™ Setting (standard or advanced)	Blend Rate (%)	Oil Specification
ISM; 1998	Standard	0.33	CG-4
ISM; 1998	Advanced	0.25	CH4, CES 20076
ISM; 1999 and later	N/A	0.33	CES 20076

Table 1, Oil Blend Rates with Centinel™ System

Engine	Centinel™ Setting (standard or advanced)	Blend Rate (%)	Oil Specification
ISM wastegated turbo	N/A	0.48	CES 20076
ISM with EGR	N/A	0.33	CI-4
L10/M11	Standard	0.33	CG-4
L10/M11	Advanced	0.25	CH4, CES 20076
N14; 1998 and earlier	Standard	0.33	CG-4
N14; 1998 and earlier	Advanced	0.25	CH-4, CES 20076
N14; 1999 and later	N/A	0.33	CES 20076
ISX with EGR	N/A	0.33	CI-4
QSX15	N/A	0.33	CES20076
K19, K38, K50	Standard	0.33	CG-4
K19, K38, K50	Advanced	0.25	CH-4, CES 20076
QST30	N/A	0.33	CG-4, CH-4, CES 20076
QSK19	Standard	0.33	CG-4, CH-4, CES 20076
QSK19	Advanced	0.25	CH-4, CES 20076
QSK45, QSK60	N/A	0.33	CG4, CH4, CES 20076

The primary recommendation is for multigrade oil of the viscosity grade indicated in Table 2. For normal ambient temperatures and operating conditions (temperatures above -15°C [5°F]), SAE 15W-40 is preferred. The use of multigrade oil reduces deposit formation, improves engine cranking in low-temperature conditions, and increases engine durability by maintaining lubrication during high-temperature operating conditions. Since multigrade oils have been shown to provide approximately 30-percent lower oil consumption, compared with monograde oils, it is important to use multigrade oils in order to make sure your engine will meet applicable emissions requirements. While the preferred viscosity grade is 15W-40, lower viscosity multigrade can be used in colder climates. (See Table 2.) Use 15W-40 when conditions permit.

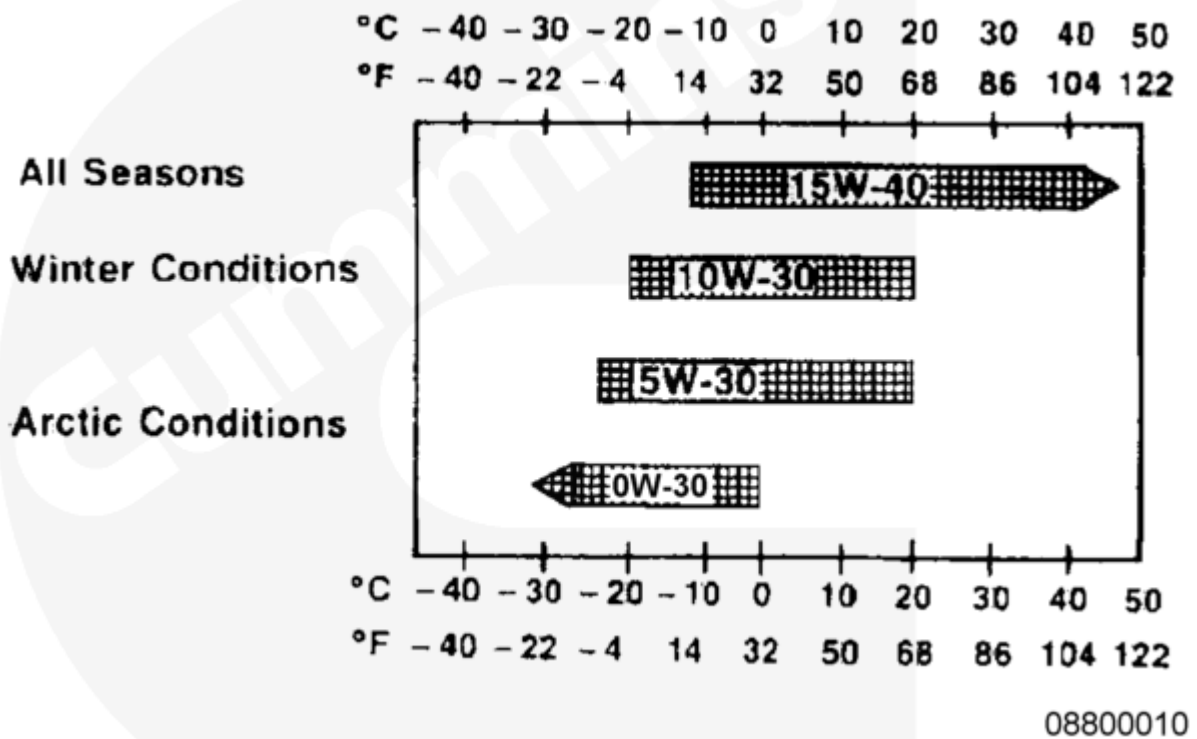


Table 2, Cummins-Recommended SAE Oil Viscosity Grades versus Ambient Temperatures

Synthetic engine oils, API category III, are recommended for use in Cummins engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Above this temperature, it is recommended that mineral-oil-based multigrade lubricants be used. Synthetic 0W-30 oils, which meet the API category III, can be used in operations where the ambient temperature never exceeds 0°C [32°F]. 0W-30 oils do **not** offer the same level of protection for fuel dilution as do the higher multigrade oils. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

Rerefined lubricating oils can be used in Cummins engines if they have an API quality designation signifying they have been tested and meet the minimum standards for that quality level. It is important to be certain these oils are actually rerefined and **not** just reclaimed. Rerefined oils have been treated to remove additives and wear metal debris, distilled and refortified with additives.

Do **not** use friction-modified oil in Cummins diesel engines unless the oil supplier can provide evidence of satisfactory performance in Cummins engines. Obtain the oil supplier's commitment that will make sure the oil is satisfactory or do **not** use that product.

